

ExifTool : A Meta-Data Extractor (For LINUX)

ABOUT EXIF

ExifTool is developed by Phil Harvey. It is a platform-independent Perl library coupled with a full-featured command-line implementation for reading, writing and manipulating the metadata across a broad range of files, particularly the JPEG images. This metadata may comprise a bunch of information such as the camera make, file type, permissions, file size etc., though it further offers more details about the photograph, like the exposure, the shutter speed and whether the flash fired or not. ExifTool probably gives us the simplest way to extract metadata from files, as it is free and an open-source program.

Installation

Exiftool is quite easy to deploy. It's just about hitting our Linux terminal and cloning the tool from GitHub.

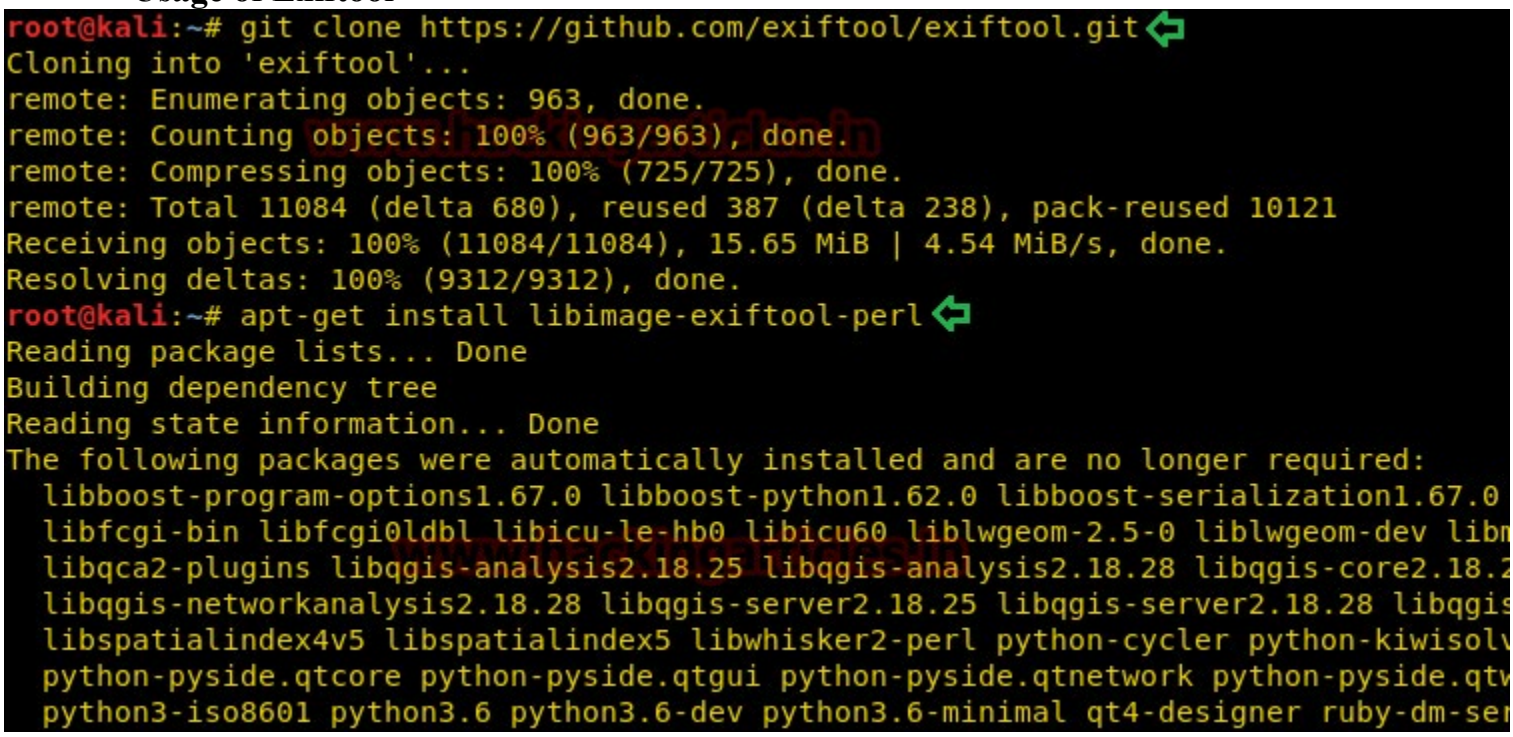
```
git clone https://github.com/exiftool/exiftool.git
```

In addition, we need to install the necessary package for it.

```
sudo apt-get install libimage-exiftool-perl
```

So, we've mounted the tool in our system. Let's take a closer look at it.

Usage of Exiftool



To extract the entire metadata of a file, we just need to execute the given below command:

```
exiftool  <filename>
```

From the below image, you can now notice that we've got all the information drawn from our image file from the very basic to advance.

```
root@kali:~# exiftool test.jpg
ExifTool Version Number         : 11.16
File Name                       : test.jpg
Directory                       : .
File Size                       : 2.3 MB
File Modification Date/Time     : 2019:07:13 12:39:53+05:30
File Access Date/Time           : 2019:07:13 12:44:30+05:30
File Inode Change Date/Time     : 2019:07:13 12:48:42+05:30
File Permissions                : rw-r--r--
File Type                       : JPEG
File Type Extension             : jpg
MIME Type                       : image/jpeg
Exif Byte Order                 : Little-endian (Intel, II)
Image Description               :
Make                            : OPPO
Camera Model Name               : OPPO F9 Pro
Orientation                     : Unknown (0)
X Resolution                    : 72
Y Resolution                    : 72
Resolution Unit                 : inches
Software                        : MediaTek Camera Application
Modify Date                     : 2019:07:13 12:34:32
Y Cb Cr Positioning             : Co-sited
Exposure Time                   : 1/33
F Number                        : 1.8
Exposure Program                : Not Defined
ISO                             : 449
Exif Version                    : 0220
Date/Time Original              : 2019:07:13 12:34:32
Create Date                     : 2019:07:13 12:34:32
Components Configuration        : Y, Cb, Cr, -
Exposure Compensation           : 0
Metering Mode                   : Center-weighted average
Light Source                    : Other
Flash                           : No Flash
Focal Length                    : 3.6 mm
Warning                         : [minor] Unrecognized MakerNotes
User Comment                    : oppo_0
```

However, if we need to capture the ids along with exif tags in the Hexa-Decimal format, though we need to run the following command:

```
exiftool -H <filename>
```

From the below image, we can see that there is a lot of information stored within these Exif tags.

```
root@kali:~# exiftool -H test.jpg
      - ExifTool Version Number         : 11.16
      - File Name                       : test.jpg
      - Directory                       : .
      - File Size                       : 2.3 MB
      - File Modification Date/Time     : 2019:07:13 12:39:53+05:30
      - File Access Date/Time           : 2019:07:13 12:52:10+05:30
      - File Inode Change Date/Time     : 2019:07:13 12:48:42+05:30
      - File Permissions                : rw-r--r--
      - File Type                       : JPEG
      - File Type Extension             : jpg
      - MIME Type                       : image/jpeg
      - Exif Byte Order                 : Little-endian (Intel, II)
0x010e Image Description                :
0x010f Make                             : OPPO
0x0110 Camera Model Name                : OPPO F9 Pro
0x0112 Orientation                      : Unknown (0)
0x011a X Resolution                     : 72
0x011b Y Resolution                     : 72
0x0128 Resolution Unit                  : inches
0x0131 Software                         : MediaTek Camera Application
0x0132 Modify Date                      : 2019:07:13 12:34:32
0x0213 Y Cb Cr Positioning              : Co-sited
```

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Extract the Common Meta-Data Information

Now execute the given below command which will provide us with the output of the most common Exif tags of the image file.

```
exiftool  -common <filename.jpg>
```

```
root@kali:~# exiftool -common test.jpg
File Name                       : test.jpg
File Size                       : 2.3 MB
Camera Model Name               : OPPO F9 Pro
Date/Time Original              : 2019:07:13 12:34:32
Image Size                      : 2128x4608
Focal Length                    : 3.6 mm
Shutter Speed                   : 1/33
Aperture                        : 1.8
ISO                             : 449
White Balance                   : Auto
Flash                           : No Flash
```

Extract the Specific Meta-Data Information

We can list a particular meta-information of our image file by simply executing the command given:

```
exiftool  -tagname  -tagname  <filename>
```

```
root@kali:~# exiftool   -FNumber -ExposureTime -ThumbnailOffset test.jpg
======== test.jpg
F Number                        : 1.8
Exposure Time                   : 1/33
Thumbnail Offset                : 1792
```

From the below image, we get our desired output displayed along with their respected tag names in a list type format.

Extract GPS Co-ordinates

The photographs we capture using our smartphones or camera have GPS coordinates embedded as metadata in the image files. To obtain this, we just need to fire the command given below:

```
exiftool  <filename> | grep GPS
```

```
root@kali:~# exiftool test.jpg | grep GPS
GPS Version ID                  : 2.2.0.0
GPS Latitude Ref                : North
GPS Longitude Ref               : East
GPS Altitude Ref                : Above Sea Level
GPS Time Stamp                  : 07:04:30
GPS Processing Method           : NETWORK
GPS Date Stamp                  : 2019:07:13
GPS Altitude                    : 0 m Above Sea Level
GPS Date/Time                   : 2019:07:13 07:04:30Z
GPS Latitude                    : 28 deg 38' 40.14" N
GPS Longitude                   : 77 deg 11' 19.06" E
GPS Position                    : 28 deg 38' 40.14" N, 77 deg 11' 19.06" E
```

Here we got the GPS Position, now just copy and paste this complete coordinate information over Google Maps and we will get the exact location of the camera when the picture was taken.

Extract Thumbnail Image

Thumbnails are the original preview images basically compressed. These are just created to open the original images more quickly and act as place holders to them. In order to extract these thumbnail images, we just need to execute the following command:

```
exiftool  -ThumbnailImage (filename) > (Output filename)
```

Here we can see that the thumbnail.jpg file is extracted from the test.jpg image.

```
root@kali:~# exiftool -ThumbnailImage test.jpg > thumbnail.jpg
root@kali:~# ls
 Backdoor_generator   Downloads
 test.html            thumbnail.jpg
 ctflab               test.jpg
```

Extract metadata using specific keywords

The following command will assist us to extract the metadata information associated with some specific keywords.

```
exiftool  "-*keyword*" <filename>
```

```
root@kali:~# exiftool "-*Image*" test.jpg
Image Description               :
Exif Image Width                : 2128
Exif Image Height               : 4608
Image Width                     : 2128
Image Height                    : 4608
Image Size                      : 2128x4608
Thumbnail Image                 : (Binary data 36864 bytes, use -b option to extract)
```

From the below picture, we can see that our fired command displays all tags with names containing the word “Image” from the file.

ExifTool’s Verbose Mode

Verbose mode generates extended information i.e. when we add [-v] to the exiftool command it will display us the comprehensive data about the process that it is performing.

```
exiftool  -v <filename>
```

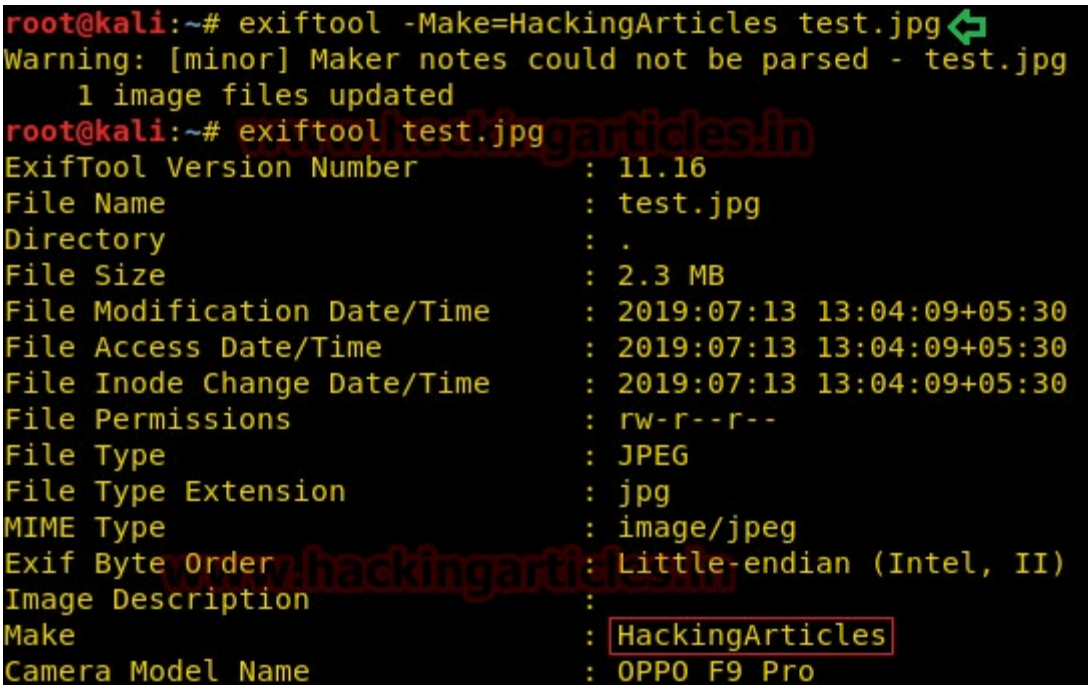
```
root@kali:~# exiftool -v test.jpg
  ExifToolVersion = 11.16
  FileName = test.jpg
  Directory = .
  FileSize = 2330562
  FileModifyDate = 1563087305
  FileAccessDate = 1563092506
  FileInodeChangeDate = 1563092487
  FilePermissions = 33188
  FileType = JPEG
  FileTypeExtension = JPG
  MIMEType = image/jpeg
JPEG APP1 (47354 bytes):
  ExifByteOrder = II
  + [IFD0 directory with 18 entries]
  | 0)  ImageDescription =
  | 1)  Make = OPPO
  | 2)  Model = OPPO F9 Pro
  | 3)  Orientation = 1
  | 4)  XResolution = 72 (72/1)
  | 5)  YResolution = 72 (72/1)
  | 6)  ResolutionUnit = 2
  | 7)  Software = MediaTek Camera Application
  | 8)  ModifyDate = 2019:07:13 12:33:21
  | 9)  YCbCrPositioning = 2
  | 10) Exif_0x0220 = 0
  | 11) Exif_0x0221 = 0
```

Writing the Meta-Data

ExifTool provides us with a great power to write most of the information on the EXIF tags, that anyone might want to alter, but some tags are protected because they describe the image's physical characteristics that we can't change with ExifTool, such as compression. Also, other tags like the GPS, the MakerNotes, this information can be edited.

To manipulate the exif data we need to execute the following command:

```
exiftool  -Make= "HackingArticles"  <filename>
```



Here we can see that the information stored in the "Make" tag is replaced from "OPPO" to "HackingArticles". While writing the information, ExifTool's script automatically preserves the original file by adding "_original" to the end of the file name.

Removing Meta-Data Information

We have only extracted or manipulated the EXIF data so far, but what if we want to remove or delete all the metadata from an image file. Just execute the following below command, let's see how this works:

```
exiftool  -all=  <filename>
```

It shows 1 image files uploaded. The "test.jpg" EXIF data has been removed effectively. Although let's attempt to extract the metadata from "test.jpg" again, hence we're just getting the basic information of the image and the rest is deleted.

```
root@kali:~# exiftool -all= test.jpg
    1 image files updated
root@kali:~# exiftool test.jpg
ExifTool Version Number         : 11.16
File Name                       : test.jpg
Directory                       : .
File Size                       : 1974 kB
File Modification Date/Time     : 2019:07:14 13:58:45+05:30
File Access Date/Time           : 2019:07:14 13:58:45+05:30
File Inode Change Date/Time     : 2019:07:14 13:58:45+05:30
File Permissions                : rw-r--r--
File Type                       : JPEG
File Type Extension             : jpg
MIME Type                       : image/jpeg
Image Width                     : 2128
Image Height                    : 4608
Encoding Process                : Baseline DCT, Huffman coding
Bits Per Sample                 : 8
Color Components                : 3
Y Cb Cr Sub Sampling            : YCbCr4:2:0 (2 2)
Image Size                      : 2128x4608
Megapixels                      : 9.8
root@kali:~# _
```

Saving outputs in Multiple Format

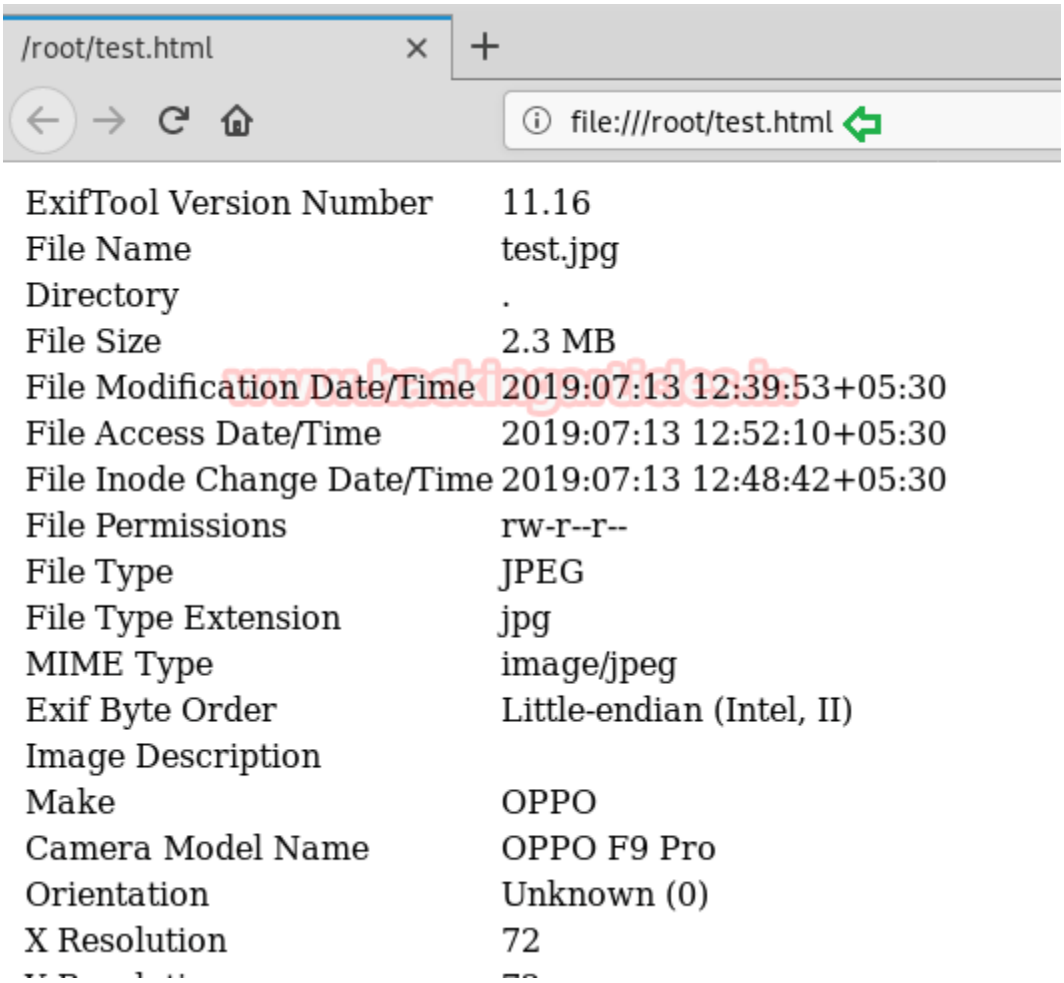
1. **In HTML file**

We will save the ExifTool's output in an HTML file in order to maintain the records and for better readability. To do this we will use the parameter "-h" along with the exiftool's command and save the results in a file with .html extension.

```
exiftool  -h (filename) > (output.html)
```

```
root@kali:~# exiftool -h test.jpg > test.html
root@kali:~# ls
 Desktop     Documents   Exploit
 ctflab      Downloads   go
 Music       test.html   hash.txt
```

Here, we can see test.html file is generated. Although we just need to open it to check our EXIF data output in any of our browsers.



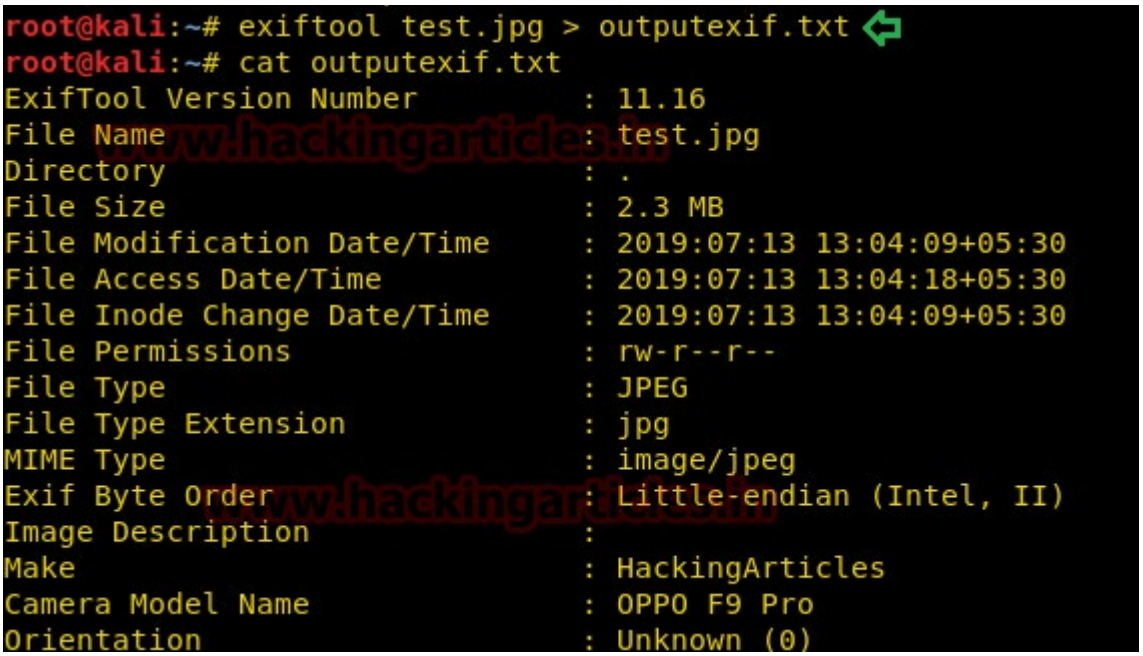
2. **In-Text File**

We can even export our exifdata to a text file similar to the output of the HTML. To achieve this, we simply need to execute the following commands:

```
exiftool (filename) > (outputexif.txt)
```

Further, we can also monitor our output either by opening it in any of the text editors or by simply running the command:

```
cat <filename>
```

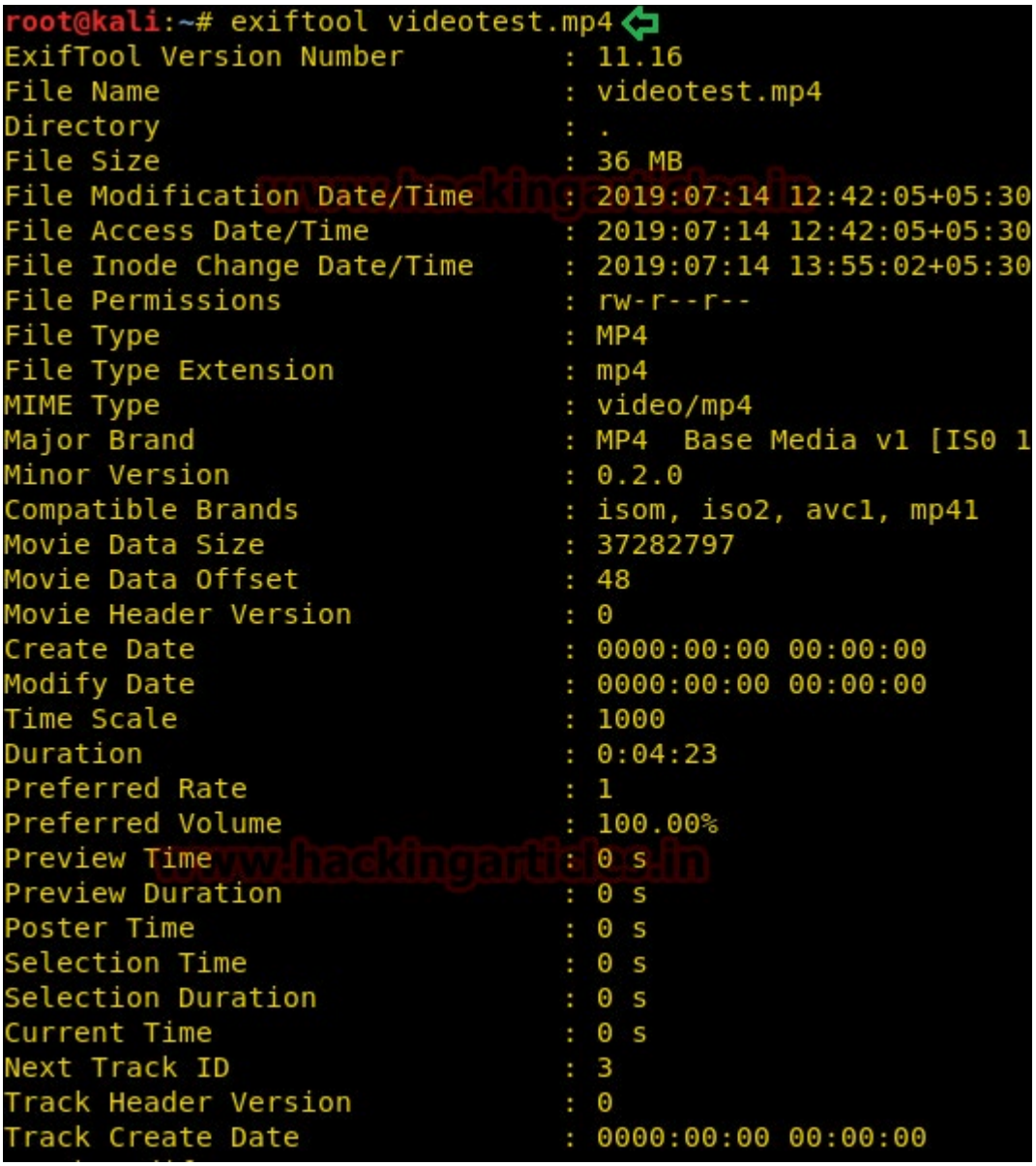


Extracting ExifData from a Video file

ExifTool not only extract metadata from the jpg file format but can also read and write in a variety of files. To know more click here.

We will now extract the entire meta-data information from an mp4 video file. To extract this, we will run the basic exiftool's command i.e.

```
exiftool <filename.mp4>
```



Conclusion

This was Exiftool's complete usability guide as a meta-data extractor. It is user-friendly and convenient because of its simple command-line implementation. It has thus become one of the best tools to extract meta-data data from a variety of file formats.